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Optimising Weak Reference Processing in the JVM Z Garbage Collector

*A Performance Analysis and Optimisation of the ZGC
Reference Processing Pipeline*



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1. Introduction

Java is one of the most widely deployed programming languages in enterprise and server-side computing [1], where applications routinely manage heaps of tens to hundreds of gigabytes. In these environments, the performance of the Java Virtual Machine’s (JVM) garbage collector (GC) directly affects application latency and throughput. The Z Garbage Collector (ZGC) addresses this by performing nearly all garbage collection work concurrently with the application, achieving millisecond pause times regardless of heap size [2].

However, one aspect of ZGC that has received comparatively little optimisation attention is its processing of *weak references*. Java’s `WeakReference` class allows applications to hold references that do not prevent the referent from being collected, and it is widely used in caching libraries, canonical mappings, and listener registrations. When the garbage collector determines that a weakly referenced object is no longer strongly reachable, it must clear the reference and, if the reference was registered with a `ReferenceQueue`, enqueue it for the application to process.

The current ZGC reference processing pipeline treats all weak references uniformly: they are linked together in an intrusive linked list, processed, and then handed to a Java-level daemon thread (the `ReferenceHandler`) for enqueueing. This design introduces overhead that is disproportionate for a specific subset of weak references: those that are not registered with a `ReferenceQueue`.

This inefficiency has been previously identified in the OpenJDK bug tracker as JDK-8029205 [3], which reports the unnecessary pending-list traversal for references without a `ReferenceQueue`. However, no implementation addressing this issue has been integrated into the JDK to date. Building on that observation, this thesis investigates whether targeted modifications to ZGC’s reference processing pipeline can reduce this overhead. Three orthogonal optimisation approaches are explored: skipping the enqueue path for references without queues, replacing the intrusive linked list with a cache-friendly dynamic array, and reducing per-reference instruction cost through an optimised clear path.

1.1 Motivation

Weak references are handled very specifically in JVM garbage collection. Unlike regular references, which are handled implicitly by the mark phase, weak

references require explicit processing by the collector. Each discovered weak reference must be individually evaluated, its referent field cleared if the referent is unreachable, and the reference potentially enqueued. This per-reference processing cost scales linearly with the number of weak references, making it a bottleneck in applications that allocate millions of them.

The problem is amplified by a mismatch between the API design and its use cases. Java's `WeakReference` API provides an optional `ReferenceQueue` parameter, and the garbage collector's pipeline is designed around the assumption that cleared references will always be enqueued. This assumption is still made even if the optional `ReferenceQueue` parameter was omitted. The result is a pipeline that performs unnecessary work for weak references that lack a `ReferenceQueue`.

ZGC's concurrent architecture makes this inefficiency particularly relevant. Because ZGC processes references concurrently with the application, any reduction in per-reference overhead translates directly into less time spent in the reference processing phase. This results in less CPU time consumed by GC threads, ultimately improving application performance.

1.2 Problem Statement

The current ZGC reference processing pipeline does not distinguish between weak references with and without a `ReferenceQueue` during its phases. All cleared references are threaded onto a pending list, transferred to the `ReferenceHandler` thread, and iterated regardless of whether they will actually be enqueued. Additionally, the intrusive linked-list data structure used for discovered references exhibits poor cache locality.

This thesis seeks to determine whether these inefficiencies can be addressed through modifications to ZGC's reference processor, and to quantify the resulting performance impact.

1.3 Research Questions

The main research questions that this thesis aims to address are:

1. What design changes and optimisations can be made to the processing of weak references without a `ReferenceQueue` in ZGC to reduce the wall-clock time spent in the reference processing phase?
2. How do these design changes and optimisations impact the wall-clock time of reference processing, the total GC cycle time, and the memory footprint (resident set size and GC-internal memory usage) of ZGC?

1.4 Stretch Goals

In addition to the primary research questions, the following stretch goals are defined to explore the broader implications of this work:

1. Does the introduction of a weak field annotation in Java provide a more efficient way to handle weak references without a `ReferenceQueue` compared to the current `WeakReference` object-based implementation?
2. Can the optimisations developed for ZGC be applied to other garbage collectors in the JVM, such as the G1 garbage collector, and if so, what performance improvements can be observed?

1.5 Scope and Delimitations

This thesis focuses on the reference processing pipeline within ZGC, specifically targeting weak references without a `ReferenceQueue`. Although the optimisation approaches are described in general terms and the stretch goals consider a wider scope, the primary implementation only targets ZGC's reference processor. Furthermore, some of the optimisations, such as the optimised clear path, may have broader applicability beyond just weak references without a `ReferenceQueue`. However, these broader applications fall outside the scope of this thesis and are not evaluated.

The correctness of the optimisations is verified through the standard OpenJDK test suite and manual inspection; formal verification is outside the scope of this work. Performance measurements are conducted on a single hardware platform under controlled conditions and therefore may not generalise to all environments.

2. Background

The three optimisation approaches introduced in Chapter 1 rely on several key concepts and design decisions in the Java Virtual Machine’s (JVM) garbage collection architecture, specifically the Z Garbage Collector (ZGC) and its reference processing pipeline. This chapter provides the necessary background on these topics to understand the motivation and implementation of the optimisations.

2.1 Garbage Collection in the JVM

The JVM provides automatic memory management through garbage collection (GC). Rather than requiring the programmer to explicitly allocate and free memory as in languages such as C or C++, the JVM tracks which objects are reachable from the application’s root set and reclaims the memory of objects that are no longer reachable. The root set consists of all data that is directly accessible to the application, such as local variables on the stack, static fields, and active threads.

The JVM ships with several garbage collector implementations, each designed with different trade-offs between throughput, latency, and memory footprint [4]. The collectors are:

- **Serial GC** A single-threaded collector that performs all garbage collection work using one thread, avoiding inter-thread communication overhead. It is best suited to single-processor systems and small heaps.
- **Parallel GC** A throughput-oriented generational collector that uses multiple GC threads to speed up collection on multiprocessor systems. It is intended for medium to large data sets.
- **G1 GC (Garbage-First)** A mostly concurrent, region-based collector designed to scale from small to large multiprocessor machines. It targets predictable pause times with high probability while maintaining high throughput and is the default collector on most platforms.
- **ZGC (Z Garbage Collector)** A scalable, low-latency collector that performs expensive GC work concurrently with application threads. It targets pause times of only a few milliseconds (largely independent of heap size) and supports heap sizes from 8 MB to 16 TB.

This thesis focuses on ZGC, as its low-latency design and concurrent reference processing pipeline can particularly benefit from optimisations that reduce the amount of work done during reference processing.

2.1.1 Garbage Collection Phases

Garbage collection generally occurs multiple times (cycles) during the lifetime of a Java application. Each cycle consists of several distinct phases, each with specific responsibilities. The exact phases and their order can vary between collectors, but a common structure is as follows:

1. **Mark Phase** The collector identifies all live objects by traversing from the root set and marking all reachable objects. Dead objects are those that are not marked. This phase may be performed concurrently with the application or during a stop-the-world pause, depending on the collector design.
2. **Reference Processing Phase** After marking completes, the collector processes *Reference* objects discovered during the mark phase. This phase determines which references have dead referents and prepares them for enqueueing. Like marking, this phase may be concurrent or done during a pause.
3. **Sweep/Reclamation Phase** The collector reclaims memory occupied by dead objects. Some collectors use a sweep algorithm (scanning the heap and freeing unmarked objects) whilst others use evacuation (copying live objects to new memory regions).
4. **Compaction/Relocation Phase** Some collectors compact the heap to eliminate fragmentation, moving live objects to consolidate free space. Others relocate objects incrementally as part of the collection process.

The order and concurrency model of these phases varies significantly between collectors. Serial and Parallel GC typically perform all phases in stop-the-world pauses. G1 performs marking and sweeping mostly concurrently with short pauses for synchronisation. ZGC performs marking, reference processing, and relocation concurrently, using barriers to maintain consistency with concurrent application threads.

2.1.2 Generational Collection

All JVM garbage collectors (Serial, Parallel, G1, and Z) employ *generational collection*, based on the *weak generational hypothesis*: most objects die young and the ones that don't tend to live for a long time [5]. To employ this, the heap is divided into at least two generations:

- The **young generation**, where newly allocated objects reside. Young generation collections are frequent but fast, since most young objects are short-lived.
- The **old generation**, where objects that have survived multiple young collections are promoted to. Old generation collections are less frequent but more expensive since live objects are more plentiful here.

ZGC adopted generational collection starting with JDK 21 [6]. In generational ZGC, the young and old generations can be collected independently and

therefore concurrently. This is relevant to reference processing because references in different generations are handled during different collection cycles.

2.2 Java Reference Types

In addition to ordinary (strong) references, Java provides *reference objects* in the `java.lang.ref` package [7] that allow applications to reference objects without preventing their garbage collection. These reference types are:

1. **SoftReference** The referent field is cleared at the discretion of the GC, typically when memory is low. Commonly used for memory-sensitive caches.
2. **WeakReference** The referent field is cleared as soon as it becomes weakly reachable (i.e., no strong or soft references exist to it). Widely used in caching frameworks, canonical mappings, and listener registrations.
3. **FinalReference** An internal JVM type (not part of the public API) used to implement the `finalize()` mechanism. Objects with finalisers are wrapped in a `FinalReference` so that the GC can schedule finalisation before reclaiming the object. This scheduling is done via a `ReferenceQueue`, which means that `FinalReferences` always have an associated `ReferenceQueue`.
4. **PhantomReference** The referent is never accessible through the reference; `get()` always returns `null`. Used for post-mortem cleanup actions, often to avoid the pitfalls of finalisation. Phantom references must be registered with a `ReferenceQueue` to be useful.

All four types extend the abstract class `Reference<T>` (in the `java.lang.ref` package), which defines the key fields of the reference object and the API for retrieving the referent, clearing the referent field, and checking if it is enqueued.

2.2.1 The Reference Object Lifecycle

Each `Reference` object maintains several fields that track its state:

- `referent` field The referenced object. The GC may clear this field (set it to `null`) when the referent becomes eligible for collection according to the reference's type.
- `queue` field The `ReferenceQueue` this reference is registered with. If the reference was not created with a queue, this field points to the sentinel object `ReferenceQueue.NULL_QUEUE`.
- `next` field Used as a link pointer when the reference is enqueued in a `ReferenceQueue`.
- `discovered` field A field with dual purpose: during GC marking, it serves as a link in the garbage collector's internal *discovered list*; after

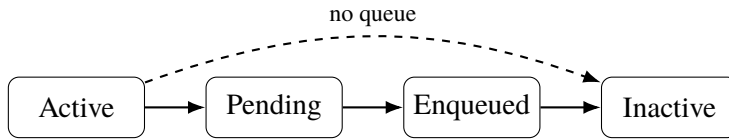


Figure 2.1. Lifecycle of `Reference` objects. References without a queue can bypass `Pending` and `Enqueued`.

processing, it is reused as a link in the VM's *pending list* that feeds the `ReferenceHandler` thread. Due to its first use, the GC can detect whether a reference has already been discovered by checking if this field is `null`.

A reference object transitions through the following states during its lifetime [8]:

1. **Active** The reference has been created and the referent is still reachable (or not yet processed).
2. **Pending** The GC has determined the referent is eligible for clearing and has added the reference to the VM's pending list. The `ReferenceHandler` thread will process it.
3. **Enqueued** The `ReferenceHandler` thread has moved the reference into its associated `ReferenceQueue`, where the application can retrieve it via `poll()` or `remove()`.
4. **Inactive** The reference has been dequeued by the application, or was created without a queue and has been cleared. It is no longer tracked by the GC or queuing infrastructure.

An important observation is that references created *without* a `ReferenceQueue` can transition directly from `Active` to `Inactive`, bypassing the `Pending` and `Enqueued` states entirely. However, as discussed in Section 2.4, the default implementation does not take advantage of this shortcut. The lifecycle and this direct shortcut path are illustrated in Figure 2.1.

2.2.2 WeakReference and ReferenceQueue

A `WeakReference` is the most commonly used non-strong reference type in Java applications. It has two constructors, as shown in Listing 1.

When a `WeakReference` is created with a `ReferenceQueue`, the application can be notified when the referent is collected. When created *without* a queue, the reference serves only as a conditional pointer, the application can call `get()` to check whether the referent is still alive, but receives no notification upon clearing.

The `ReferenceQueue` is implemented as a synchronized singly-linked list using the `next` field of the `Reference` objects. It provides `poll()` for non-blocking retrieval and `remove()` for blocking retrieval with an optional

```

1 // Without a queue - queue field set to NULL_QUEUE
2 public WeakReference(T referent) {
3     super(referent);
4 }
5
6 // With a queue
7 public WeakReference(T referent, ReferenceQueue<? super
8     T> q) {
9     super(referent, q);
10 }

```

Listing 1: WeakReference constructors

timeout. This is the mechanism by which applications can react to the collection of referents, for example to clean up associated resources.

A key insight for this thesis is that many real-world uses of `WeakReference` do not require a `ReferenceQueue`. Caching frameworks, `WeakHashMap`, and similar data structures typically create weak references without queues, relying instead on periodic polling or lazy cleanup. This means the garbage collector expends effort preparing these references for enqueueing (adding them to the pending list) even though the `ReferenceHandler` thread will immediately skip them.

2.3 The Z Garbage Collector

ZGC is a concurrent, region-based, compacting garbage collector included in the JDK since JDK 11 (experimental) [2] and production-ready since JDK 15 [9]. Starting with JDK 21, ZGC supports generational collection, which is now the default mode [6]. ZGC’s primary design goals are [2]:

- Sub-millisecond pause times that do not increase with heap size.
- Support for heap sizes from 8 MB to 16 TB.
- Concurrent marking, relocation, and reference processing.

ZGC achieves these goals through several key techniques:

Coloured Pointers.

ZGC embeds metadata in the unused bits of 64-bit object pointers. These *colour bits* encode information about the pointer’s current state relative to the GC’s phase, for example, whether the pointer has been remapped after relocation, or whether it points to a young or old generation object. This allows the GC to reason about pointer states without loading the object header.

Barriers.

ZGC uses barriers at every reference load or write. The barrier checks the colour bits of the loaded pointer and, if necessary, applies a fixup (e.g., remap-

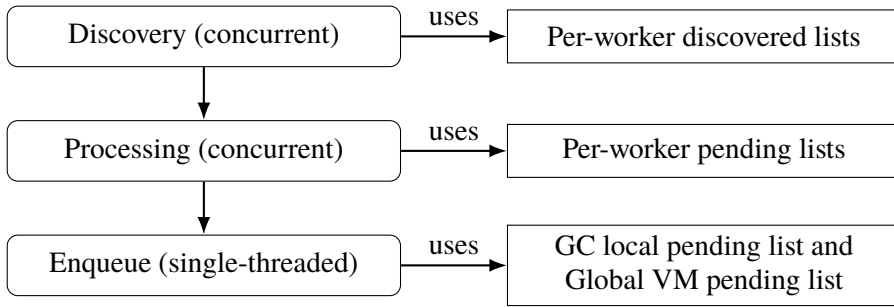


Figure 2.2. ZGC reference-processing pipeline and data flow from discovery to Java-level enqueue handling.

ping after relocation). This is the mechanism that allows ZGC to perform most of its work concurrently with the application.

Concurrent Phases.

A ZGC collection cycle interleaves short stop-the-world pauses (for root scanning and synchronisation) with concurrent phases (for marking, reference processing, and relocation). The concurrent phases run alongside application threads, using barriers to maintain consistency.

Multiple GC Worker Threads.

ZGC uses multiple GC worker threads to parallelize GC work. Each worker thread maintains its own data structures in many cases to avoid contention.

2.4 Reference Processing Pipeline in ZGC

Reference processing is the mechanism by which the garbage collector handles `Reference` objects discovered during marking. The pipeline has three stages: *discovery*, *processing*, and *enqueueing*. In ZGC, the first two stages run concurrently with the application during the marking phase, while the enqueue stage involves handing off references to the Java-level `ReferenceHandler` thread. An overview of this data flow is shown in Figure 2.2.

2.4.1 Discovery

During the concurrent marking phase, when ZGC encounters a `Reference` object (an instance of `SoftReference`, `WeakReference`, `FinalReference`, or `PhantomReference`), it evaluates whether the reference should be *discovered*, i.e. added to a list for later processing.

The discovery decision depends on several factors:

1. **Is the reference active?** Only active references are eligible. For non-`FinalReference` types, an inactive reference has a `null` referent field. For `FinalReference`, inactivity is indicated by a non-null `next` field.
2. **Is the referent still strongly reachable?** If the referent is already marked as strongly live, the reference does not need processing. The garbage collector checks this via its liveness information.
3. **Soft reference policy.** For `SoftReference` objects, the collector applies a policy (based on available memory and the reference's last access time) to decide whether to keep the referent alive or clear the referent field.
4. **Generational considerations.** In ZGC, only references residing in the old generation are discovered; young-generation references are always treated as strong. This is because reference processing introduces additional delay and uncertainty to the duration of the young-generation collection. By treating young references as strong, ZGC can complete young collections more predictably, which decreases the risk of the application running out of memory during a long young collection.

Once a reference is selected for discovery, it is recorded on a *discovered list*. In ZGC, each GC worker thread maintains its own discovered-list head, so discovery is implemented as a cheap, non-atomic prepend to a per-worker list. This avoids CAS contention found in shared processors where multiple threads update the same discovered head.

2.4.2 Processing

After marking completes, the discovered lists are processed. For each discovered reference, the collector determines the final fate of the referent:

1. If the referent field has become `null` (cleared by the application concurrently), the reference is dropped from the list.
2. If the referent is now strongly reachable (e.g. it was marked as alive after the reference's discovery), the reference is dropped.
3. If the referent is still unreachable according to the reference's strength, the referent field is **cleared** (set to `null`), and the reference is prepared for enqueueing.

In ZGC, clearing involves setting the `referent` field to a coloured null pointer via compare-and-swap (CAS) to ensure thread safety since application threads may be concurrently accessing the reference.

After clearing, the reference is linked onto the thread-local pending list. This involves writing the reference's address into the `discovered` field of the previously processed reference. Once the thread has processed all references in its discovered list, the thread-local list is atomically prepended to ZGC's internal pending list.

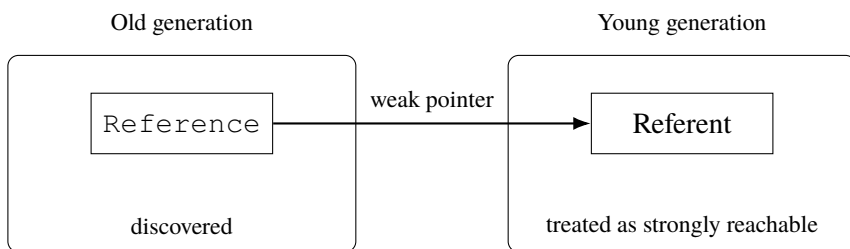


Figure 2.3. If an old-generation `Reference` points to a young referent, the young referent must remain alive during young collection.

Generational interactions are important during processing: when a `Reference` object is old but its referent is young, the referent is treated as strongly reachable and the referent field must not be cleared. Processing must therefore verify generation membership and, if appropriate, ensure young-generation liveness is preserved (by marking the young referent), as illustrated in Figure 2.3.

2.4.3 Enqueue

After a reference's referent field has been cleared, the reference must be made available to the application if it was created with a `ReferenceQueue`. This happens through the *pending list* mechanism which feeds the `ReferenceHandler` thread. The handoff process is as follows:

1. The GC has built an internal pending list by linking cleared references together via their `discovered` field.
2. Under the heap lock, the GC atomically prepends its internal pending list onto the global VM pending list.
3. The `ReferenceHandler` daemon thread is notified.
4. The `ReferenceHandler` thread wakes up, atomically retrieves the pending list, and iterates over it. For each reference, it checks whether `queue != NULL_QUEUE`. If the reference has a real queue, it calls `queue.enqueue(this)`; otherwise it simply advances to the next reference.

This handoff reveals an inefficiency: references created without a `ReferenceQueue` are still linked into the pending list, transferred to the `ReferenceHandler` thread, and iterated over only to be skipped. For workloads where the majority of weak references lack queues, this results in a considerable amount of extra work. This behaviour has been reported in an OpenJDK enhancement issue [3]. Avoiding enqueue-path work for references with `NULL_QUEUE` is one of the primary optimisation targets explored in this thesis.

3. Method

In order to answer the research questions posed in Section 1.3, three orthogonal optimisation approaches were implemented separately for the processing of weak references without a `ReferenceQueue` in the Z garbage collector (ZGC). These separate implementations were then combined in all possible combinations and had their performance evaluated using microbenchmarks. The three approaches were, as introduced in Chapter 1, the following:

- Skipping the enqueue path for weak references without a `ReferenceQueue`
- Replacing the intrusive linked list with a dynamic array for the discovered list
- Optimised Clear Path

The rationale, implementation details, and performance testing methodology of these approaches are described in the following sections. The evaluation of the approaches is described in Chapter 4. The complete changed source code for the implemented optimisations is provided in Appendix A.

3.1 Skipping Enqueue Path

As described in Section 2.4.2, ZGC enqueues discovered references to the internal pending list regardless of whether they have a `ReferenceQueue` or not. This is inefficient for weak references without a `ReferenceQueue` as discussed in Section 2.4.3. Because of this, the reference processing pipeline was changed so that the enqueue path can be skipped for weak references without a `ReferenceQueue`. The intended effect of this optimisation is to both reduce the amount of work done by the GC threads during reference processing and to reduce the amount of work done by the `ReferenceHandler` thread when iterating over the pending list.

3.1.1 Implementation of Null-queue Fast Path

In order to skip enqueueing weak references without a `ReferenceQueue` the `ReferenceProcessor` class was modified to check if the queue field of a weak reference being discovered is equal to the sentinel value `ReferenceQueue.NULL_QUEUE` before adding it to the discovered list. If the queue field is equal to `ReferenceQueue.NULL_QUEUE` (the weak reference was

```

1 bool discover_reference(oop ref_obj, ReferenceType type)
2 {
3     zaddress ref = to_zaddress(ref_obj);
4
5     zpointer* referent_field = reference_referent_addr(
6         ref);
7     zaddress referent = load_barrier(referent_field);
8     if (!should_discover(ref, type, referent)) {
9         return false;
10    }
11
12    if (type == REF_WEAK && !has_reference_queue(ref)) {
13        weak_no_queue_list_per_worker.append(ref);
14    } else {
15        discovered_list_per_worker.append(ref);
16    }
17
18    mark_discovered(ref);
19    return true;
20 }

```

Listing 2: Pseudocode of the null-queue fast path during discovery. Queue-less weak references are routed to a separate discovered list and processed by a no-enqueue pipeline. The corresponding implementation excerpts are provided in Appendix A.

not created with a queue), it is added to a separate discovered list for weak references without a `ReferenceQueue` instead of being added to the regular discovered list. This separate discovered list is then processed independently by the GC threads and none of its weak references are enqueued to the pending list for the `ReferenceHandler` thread. Each GC thread maintains its own separate discovered list for weak references without a `ReferenceQueue` to avoid contention between threads.

The reason for separating the discovered lists instead of simply skipping enqueueing for weak references without a `ReferenceQueue` is that two unique reference processing functions can then be used for the two lists: one that includes enqueueing and one that does not. This allows for more efficient processing of the weak references without a `ReferenceQueue`, as it avoids unnecessary checks for whether the weak reference has a queue during processing.

This discovery-time split is summarised in Listing 2.

3.2 Replacing Intrusive Linked List with Dynamic Array

When processing the discovered references, ZGC iterates over the discovered list. This list is, as described in Section 2.4.1, implemented as an intrusive linked list of the discovered references. This iteration over a linked list is inefficient since it has poor cache locality [10] and cannot exploit instruction-level parallelism between iterations. Because of this, the intrusive linked list was replaced with a dynamic array for the discovered list for weak references without a `ReferenceQueue`. The intended effect of this optimisation is to reduce the amount of time spent loading the next reference to process and thereby allow for more efficient processing of the discovered references through better cache locality and better exploitation of instruction-level parallelism.

3.2.1 Implementation of Dynamic Array for Discovered References

The implemented dynamic array, called `ZAddressArray`, stores discovered references in a contiguous layout rather than linking them via the `discovered` field. The array is allocated on the C heap and grows using a power-of-two strategy (with a minimum capacity of 8), using `memcpy` for bulk copying during expansion rather than element-by-element loops.

Each GC worker thread maintains its own `ZAddressArray` instance to avoid contention. During discovery, when a reference should be added to a discovery list, its data is appended to the worker's array instead of being appended to a linked list. The reference's `discovered` field is still set (to a self-referencing pointer) to mark the reference as discovered, preventing duplicate discovery.

During processing, the array is iterated sequentially using index-based access. Because the elements are stored contiguously in memory, this iteration benefits from hardware prefetching and improved cache line utilisation compared to following pointers scattered across the heap.

When using both dynamic arrays and separate discovered lists, only the discovered list for weak references without a `ReferenceQueue` is backed by a `ZAddressArray`, while the regular discovered list continues to use the linked list.

The resulting discovery and processing flow is summarised in Listing 3.

3.3 Optimised Clear Path

The standard reference processing path in ZGC uses general-purpose functions designed to handle all reference types uniformly. This generality introduces


```

1 void discover_reference(oop ref_obj, ReferenceType type)
2 {
3     //...
4     ZWeakRefData data;
5     data.set_from_ref(ref);
6     worker_array.append(data)
7     //...
8 }
9 void process_worker_discovered_weak_refs_without_queue(
10     ZAddressArray& arr) {
11     size_t dropped = 0;
12     for (size_t i = 0; i < arr.length(); i++) {
13         const ZWeakRefData& data = arr.at(i);
14         data.mark_not_discovered();
15
16         if (try_make_inactive(data)) {
17             cleared_weak_no_queue_count++;
18         } else {
19             dropped++;
20         }
21     }
22     arr.clear_and_reserve(dropped);
23 }

```

Listing 3: Pseudocode for the ZAddressArray-backed queue-less weak-reference path. Discovery appends pre-fetched fields and processing scans the array sequentially. The corresponding source code is provided in Appendix A.

per-reference overhead that is unnecessary when processing only weak references without a `ReferenceQueue`, where the reference type is always `REF_WEAK` and enqueueing is never required.

The standard function for determining whether a reference's referent should be cleared performs several operations per reference:

1. A load barrier to read the referent from the reference object.
2. A virtual call to determine the reference type (soft, weak, final, or phantom).
3. A compare-and-swap (CAS) operation via a ZGC barrier to atomically set the referent field to a coloured null pointer.

For weak references without a `ReferenceQueue`, each of these operations can be simplified or eliminated:

1. When combined with the dynamic array optimisation, the referent field address and its value can be loaded during discovery and stored in the array entry, avoiding redundant memory loads during processing. This pre-loading is safe as long as the referent field is not set to a non-null value without any safeguards during processing. Since the referent can only be concurrently modified to null, which happens when an application thread clears the reference.
2. The reference type is known to be `REF_WEAK` at the call site, eliminating the need for the virtual call to determine the type.
3. Since the only concurrent modification to the referent field that can occur is from an application thread setting the referent to null. The CAS operation to be replaced with a simple store of a coloured null pointer without needing to check for concurrent updates.

3.3.1 Implementation of Optimised Clear Path

The implementation introduces a new function for determining whether a queue-less weak reference's referent should be cleared. This function takes a data struct (either from the dynamic array or constructed on the fly from the linked list) containing the pre-fetched field addresses and values.

The new function checks whether the referent is strongly live via metadata in ZGC's heap management. If the referent is not strongly live and resides in the old generation, its referent field is cleared using a direct pointer store rather than a CAS. This is safe since, as mentioned previously, the only possible new value is null so the worst-case scenario is that null is overwritten with null. If the referent is still live (either strongly marked or implicitly by residing in the young generation), the referent field is recoloured with the correct pointer metadata bits, this recolouring, however, has to occur with a CAS so that a null value is not overwritten with a non-null value. Additionally, if the referent is young and the young generation is currently in its mark phase, the referent is marked to preserve cross-generational liveness.

The discovered field is also cleared using a direct store to the field address rather than through an atomic access function, since no other thread modifies this field after discovery.

This specialised inactive-check path is summarised in Listing 4.

3.4 Build Configurations

The three optimisation approaches are controlled by compile-time preprocessor flags: `ZUseOptimisedClearPath` for the optimised clear path, `ZUseSeparatedDiscoveredLists` for the separate discovered list, and `ZUseDynamicArray` for the dynamic array. Each flag can be independently enabled (1) or disabled (0) via compiler flags passed during the JDK build.

Eight build configurations were created to evaluate all combinations of the three optimisation flags, as shown in Table 3.1. Each configuration produces a separate JDK build with its own binary, allowing direct comparison under identical benchmark conditions.

Table 3.1. *Build configurations and their enabled optimisation flags*

Configuration	Clear Path	Sep. Lists	Dyn. Array
none	—	—	—
dyn_only	—	—	✓
sep_only	—	✓	—
sep_dyn	—	✓	✓
clear_path_only	✓	—	—
clear_path_dyn	✓	—	✓
clear_path_sep	✓	✓	—
all	✓	✓	✓

3.5 Performance Evaluation

To evaluate the performance impact of the optimisation approaches, two microbenchmarks were developed that exercise the weak reference processing pipeline under controlled conditions. The benchmarks and their execution environment are described below.

3.5.1 Benchmark Design

Two synthetic microbenchmarks were developed to evaluate the optimisation approaches. The first benchmark isolates the reference processing cost, and the second is a slightly more realistic benchmark that introduces additional variability.

```

1 bool try_make_inactive_fast(const ZWeakRefData& data) {
2     // if used with the dynamic array
3     zaddress referent = data.referent_addr;
4
5     // if used with the linked list
6     zaddress referent = load_barrier(data.
7         referent_field_addr);
8
9     // Young referents are implicitly kept alive by
10    generational rules.
11    if (is_young(referent)) {
12        if (young_mark_is_active()) {
13            mark_young_root(referent);
14        }
15        return false; // keep as active
16    }
17
18    // Old referent: clear only if not strongly live.
19    if (!is_strongly_live(referent)) {
20        *data.referent_field_addr = to_zpointer(zaddress
21            ::null);
22        return true; // became inactive
23    }
24
25    // Still live: recolour pointer metadata, but do not
26    overwrite null.
27    zpointer observed = data.referent_field_value;
28    zpointer recoloured = remap_with_good_metadata(
29        referent);
30    cas_referent_if_non_null(data.referent_field_addr,
31        observed, recoloured);
32    return false;
33 }

```

Listing 4: Pseudocode for the specialised queue-less weak-reference inactive check. The old-generation clear path uses direct stores, while recolouring uses CAS to avoid null-to-non-null races. The corresponding implementation excerpts are provided in Appendix A.

Single-Object Benchmark

The Single-Object benchmark (`WeakRefSingleObjectBenchmark`) allocates 10 million `WeakReference` objects that all point to a single target object. The weak references are stored in a shuffled order to reflect realistic allocation patterns where references and other objects are interleaved in memory. None of the weak references are created with a `ReferenceQueue`. After a brief holding period, the single strong reference to the target object is released and a GC cycle is triggered manually. Because all weak references become clearable simultaneously in a single GC cycle, this benchmark maximises the per-cycle reference processing load and isolates the cost of the processing pipeline itself.

Multi-Object Benchmark

The multi-object benchmark (`WeakRefMultiObjectBenchmark`) is a slightly more realistic workload. It allocates approximately 7.3 million objects, each containing a randomly sized byte array payload (between 509 and 1097 bytes). Each object is held by both a strong reference and a `WeakReference`. Of the weak references, 5% are created with a `ReferenceQueue` and 95% without, reflecting a workload dominated by queue-less weak references but where the enqueue path is still exercised for a minority of references. The allocation order of both strong and weak references is randomised using a Fisher-Yates shuffle [11] to avoid artificial locality patterns.

Unlike the synthetic benchmark, the strong references are not all released at once. Instead, the benchmark proceeds through five rounds of clearing. In each round, approximately 20% of the strong references are nulled in a randomised order, making the corresponding objects eligible for garbage collection. Between rounds, the benchmark sleeps to allow GC cycles to discover and process the weak references whose referent now is unreachable. This gradual release pattern means that weak references are discovered and processed across multiple GC cycles rather than in a single burst.

The benchmark runs three iterations with a five-second cooldown between them to allow the JVM to stabilise. Each iteration performs the full allocation, clearing, and GC cycle sequence independently.

3.5.2 Execution Environment

To ensure reproducible measurements, the benchmarks were executed under controlled conditions using a dedicated runner script. The script applies the following isolation measures:

- **CPU pinning:** The Java process was pinned to a fixed set of CPU cores using `taskset`. Meanwhile, the monitoring and scripting processes ran on separate cores to avoid interference.
- **CPU frequency control:** The CPU governor is set to performance mode at the base clock frequency. Both the minimum and maximum scaling

frequencies are locked to the base frequency to prevent frequency scaling from introducing variance.

- **Process priority:** The Java process is run with elevated priority (`nice -n -5`) to reduce scheduling interference from other system processes.
- **Cache clearing:** Filesystem caches are dropped between runs by writing to `/proc/sys/vm/drop_caches`.
- **Cooldown period:** A configurable cooldown period (default 10 seconds) is inserted between variant runs to allow the system to return to a stable state.

For the single-object benchmark, the JVM was configured with a fixed 7 GB heap and ZGC enabled. Tenuring thresholds are set to 1 to promote the target object to the old generation quickly, ensuring that the reference processor discovers it. GC statistics and reference processing logs were enabled to collect the results from.

For the multi-object benchmark the JVM was configured with a fixed 8 GB heap, ZGC enabled, and a forced major collection interval of 0.5 seconds to ensure frequent GC activity during the benchmark. Tenuring thresholds are set to 1 to promote objects to the old generation quickly, ensuring that the reference processor discovers them. This is necessary since ZGC doesn't perform reference processing in the young generation. GC statistics and reference processing logs were enabled to collect the results from.

3.5.3 Measurement Methodology

Each benchmark was run for 100 iterations. Within each iteration, all eight variants are run in sequence with environment preparation between each. This interleaved ordering mitigates systematic drift in system conditions across the measurements affecting one variant more than others.

During each run, a memory monitoring script samples memory metrics from the process at high frequency. Additionally, the current benchmark phase is tracked and for the multi-object benchmark the inner iteration is tracked as well. This tracking is done from the benchmark's log file to enable phase-aligned analysis. See Table 3.2 for the memory metrics collected.

In addition to the continuous memory samples, GC statistics are parsed from the log files of each run log. For each GC metric, the parser extracts the Total Avg/Max pair and unit reported by ZGC. For the single-object benchmark, Max is used as the primary value for comparison; for the multi-object benchmark, Avg is used. See Table 3.3 for the GC log metrics collected.

Table 3.2. *Continuous monitoring metrics used in the measurement pipeline.*

Metric	Units	Description
Timestamp	ms	Sample timestamp used to align and aggregate runs over time.
RSS	kB	Process resident set size (RSS), representing total physical memory in use by the JVM process.
GC Committed Memory	kB	GC-related memory currently committed by the JVM (actually backed memory from Native Memory Tracking).
Java Heap	kB	Java heap usage sampled from <code>jcmd GC.heap_info</code> .
Phase	label	Benchmark phase label emitted by the benchmark and used for phase-aligned analysis.
Iteration	label	Inner-iteration identifier (multi-object benchmark) used to align repeated phase sequences.

Table 3.3. *GC log metrics used in the measurement pipeline.*

Metric	Units	Description
Major Collection	ms	End-to-end duration of major collections reported by GC statistics.
Old Generation	ms	Total time spent in old-generation collection work.
Old Subphase: Concurrent Mark Follow	ms	Time spent in old-generation concurrent mark-follow processing.
Old Subphase: Concurrent References Process	ms	Time spent in concurrent reference processing in old-generation collection.
Young Subphase: Concurrent Mark Follow	ms	Time spent in young-generation concurrent mark-follow processing.

4. Results

In order to evaluate the effectiveness of the implemented optimisations, the two synthetic benchmarks described in Section 3.5.1 were executed as outlined in Section 3.5.2 and the metrics listed in Section 3.5.3 were collected.

The memory metrics, namely the average of the resident set size (RSS), GC committed memory, and Java heap size were measured and compared between the baseline and optimised implementations. The results of these measurements are presented in ...

The performance metrics, namely the average duration of major collections, old generation collections, old generation mark and follow, old generation reference processing, and young generation mark and follow were also measured and compared between the baseline and optimised implementations. The results of these measurements are presented in ...

5. Discussion

6. Conclusion

References

- [1] Wikipedia, “Programming languages used in most popular websites.” https://en.wikipedia.org/wiki/Programming_languages_used_in_most_popular_websites, 2025. Accessed: 2026-03-12.
- [2] OpenJDK, “Jep 333: Zgc: A scalable low-latency garbage collector (experimental).” <https://openjdk.org/jeps/333>, 2018. Accessed: 2026-03-12.
- [3] OpenJDK Bug System, “Unnecessary pending-list traversal for references without referencequeue.” <https://bugs.openjdk.org/browse/JDK-8029205>, n.d. Accessed: 2026-02-24.
- [4] Oracle, “Available collectors.” <https://docs.oracle.com/en/java/javase/25/gctuning/available-collectors.html>, 2025. Java Platform, Standard Edition 25 Garbage Collection Tuning Guide. Accessed: 2026-02-24.
- [5] H. Lieberman and C. Hewitt, “A real time garbage collector based on the lifetimes of objects,” Tech. Rep. AIM-569a, MIT Artificial Intelligence Laboratory, Oct. 1981. Accessed: 2026-02-24.
- [6] OpenJDK, “Jep 439: Generational zgc.” <https://openjdk.org/jeps/439>, 2023. Accessed: 2026-03-12.
- [7] Oracle, “java.lang.ref (java se 25).” <https://docs.oracle.com/en/java/javase/25/docs/api/java.base/java/lang/ref/package-summary.html>, 2025. Accessed: 2026-03-12.
- [8] Oracle, “Reference (java se 25).” <https://docs.oracle.com/en/java/javase/25/docs/api/java.base/java/lang/ref/Reference.html>, 2025. Accessed: 2026-03-12.
- [9] OpenJDK, “Jep 377: Zgc: A scalable low-latency garbage collector (production).” <https://openjdk.org/jeps/377>, 2020. Accessed: 2026-03-12.
- [10] T. M. Chilimbi, B. Davidson, and J. R. Larus, “Cache-conscious structure layout,” in *Proceedings of the ACM SIGPLAN 1999 Conference on Programming Language Design and Implementation (PLDI)*, pp. 1–12, ACM, 1999.
- [11] D. E. Knuth, *The Art of Computer Programming, Volume 2: Seminumerical Algorithms*. Addison-Wesley, 3rd ed., 1997. Section 3.4.2, Algorithm P.

Appendix A.

Changed Source Code

This appendix lists the complete source files changed to implement the weak-reference processing optimisations evaluated in this thesis. For readability and to keep the thesis build stable, representative changed regions are included below.

Full changed source files:

- `src/hotspot/share/gc/z/zConfFlags.h`
- `src/hotspot/share/gc/z/zAddressArray.hpp`
- `src/hotspot/share/gc/z/zReferenceProcessor.hpp`
- `src/hotspot/share/gc/z/zReferenceProcessor.cpp`

Configuration Flags

File: `src/hotspot/share/gc/z/zConfFlags.h`

```
1 #ifndef ZUseOptimisedClearPath
2 #define ZUseOptimisedClearPath 1
3 #endif // ZUseOptimisedClearPath
4
5 #ifndef ZUseSeparateDiscoveredLists
6 #define ZUseSeparateDiscoveredLists 1
7 #endif // ZUseSeparateDiscoveredLists
8
9 #ifndef ZUseDynamicArray
10 #define ZUseDynamicArray 1
11 #endif // ZUseDynamicArray
```

Dynamic-array Data Structure

File: `src/hotspot/share/gc/z/zAddressArray.hpp`

```
1 /*
2  * Copyright (c) 2025, Oracle and/or its affiliates. All
3  * rights reserved.
4  * DO NOT ALTER OR REMOVE COPYRIGHT NOTICES OR THIS FILE
5  * HEADER.
```

```

4  *
5  * This code is free software; you can redistribute it
   and/or modify it
6  * under the terms of the GNU General Public License
   version 2 only, as
7  * published by the Free Software Foundation.
8  *
9  * This code is distributed in the hope that it will be
   useful, but WITHOUT
10 * ANY WARRANTY; without even the implied warranty of
   MERCHANTABILITY or
11 * FITNESS FOR A PARTICULAR PURPOSE. See the GNU
   General Public License
12 * version 2 for more details (a copy is included in the
   LICENSE file that
13 * accompanied this code).
14 *
15 * You should have received a copy of the GNU General
   Public License version
16 * 2 along with this work; if not, write to the Free
   Software Foundation,
17 * Inc., 51 Franklin St, Fifth Floor, Boston, MA
   02110-1301 USA.
18 *
19 * Please contact Oracle, 500 Oracle Parkway, Redwood
   Shores, CA 94065 USA
20 * or visit www.oracle.com if you need additional
   information or have any
21 * questions.
22 */
23
24 #ifndef SHARE_GC_Z_ZADDRESSARRAY_HPP
25 #define SHARE_GC_Z_ZADDRESSARRAY_HPP
26
27 #include "gc/z/zAddress.hpp"
28 #include "memory/allocation.hpp"
29 #include "utilities/debug.hpp"
30 #include "utilities/powerOfTwo.hpp"
31 #include "utilities/globalDefinitions.hpp"
32 #include <string.h>
33 #include "gc/z/zConfFlags.h"
34
35 struct ZWeakRefData {
36 #if ZUseOptimisedClearPath
37     zpointer* referent_field_addr;
38     address* discovered_field_addr;
39     address referent_addr;

```

```

40     zpointer referent_field_value;
41 #if !ZUseSeparateDiscoveredLists
42     // For the case where we use a dynamic array but not
        separate discovered lists, we need to store the
        reference itself in the data struct to be able to
        mark it as discovered.
43     zaddress reference;
44 #endif // !ZUseSeparateDiscoveredLists
45 #else //!ZUseOptimisedClearPath
46     zaddress reference;
47 #endif // ZUseOptimisedClearPath
48
49 };
50
51 // High-performance growable array specifically for
        storing discovered weak references.
52 // Uses array-of-structs (AoS) layout for better cache
        locality when processing sequentially.
53 // Each element is a ZWeakRefData struct containing:
54 // - referent_field_addr: pointer to zpointer field (the
        referent field in Reference object)
55 // - discovered_field_addr: pointer to zaddress field (
        the discovered field in Reference object)
56 // - referent_addr: zaddress value of the referent
        object
57 //
58 // Performance optimizations:
59 // - Uses memcpy for bulk copying instead of element-by-
        element loops
60 // - Does not zero newly allocated memory (caller
        responsible for initialization)
61 // - Single clear_and_reserve operation to avoid
        separate clear/reserve calls
62 // - Array-of-structs layout improves cache efficiency
        for sequential processing
63 class ZAddressArray : public AnyObj {
64 private:
65     ZWeakRefData* _data;
66     size_t         _length;
67     size_t         _capacity;
68
69     void expand_to(size_t new_capacity) {
70         assert(new_capacity > _capacity, "expected growth
            but %d <= %d", new_capacity, _capacity);
71
72         ZWeakRefData* new_data = (ZWeakRefData*)AllocateHeap
            (new_capacity * sizeof(ZWeakRefData), mtGC);

```

```

73
74     if (_length > 0) {
75         // Use memcpy for trivially copyable types - much
76         // faster than element-by-element copy
77         memcpy(new_data, _data, _length * sizeof(
78             ZWeakRefData));
79     }
80
81     if (_data != nullptr) {
82         FreeHeap(_data);
83     }
84
85     _data = new_data;
86     _capacity = new_capacity;
87 }
88
89 void grow(size_t min_capacity) {
90     size_t new_capacity = MAX2((size_t) 8,
91         next_power_of_2(min_capacity - 1));
92     expand_to(new_capacity);
93 }
94
95 public:
96     ZAddressArray() : _data(nullptr), _length(0),
97         _capacity(0) {}
98
99     ~ZAddressArray() {
100         if (_data != nullptr) {
101             FreeHeap(_data);
102             _data = nullptr;
103         }
104     }
105
106     // Append a new entry
107 #if ZUseOptimisedClearPath &&
108     ZUseSeparateDiscoveredLists
109 void append(zpointer* referent_field_addr, zaddress*
110     discovered_field_addr, zaddress referent_addr,
111     zpointer referent_field_value) {
112     if (_length >= _capacity) {
113         grow(_length + 1);
114     }
115     _data[_length].referent_field_addr =
116         referent_field_addr;
117     _data[_length].discovered_field_addr =
118         discovered_field_addr;
119     _data[_length].referent_addr = referent_addr;

```

```

111     _data[_length].referent_field_value =
112         referent_field_value;
113 }
114 #elif ZUseOptimisedClearPath && !
115     ZUseSeparateDiscoveredLists
116 void append(zpointer* referent_field_addr, zaddress*
117     discovered_field_addr, zaddress referent_addr,
118     zpointer referent_field_value, zaddress reference)
119 {
120     if (_length >= _capacity) {
121         grow(_length + 1);
122     }
123     _data[_length].referent_field_addr =
124         referent_field_addr;
125     _data[_length].discovered_field_addr =
126         discovered_field_addr;
127     _data[_length].referent_addr = referent_addr;
128     _data[_length].referent_field_value =
129         referent_field_value;
130     _data[_length].reference = reference;
131     _length++;
132 }
133 #else //!ZUseOptimisedClearPath
134 void append(zaddress reference) {
135     if (_length >= _capacity) {
136         grow(_length + 1);
137     }
138     _data[_length].reference = reference;
139     _length++;
140 }
141 #endif // ZUseOptimisedClearPath
142
143 // Get entry at index
144 const ZWeakRefData& at(size_t index) const {
145     assert(index >= 0 && index < _length, "index out of
146         bounds: %d (length: %d)", index, _length);
147     return _data[index];
148 }
149
150 // Current length
151 size_t length() const {
152     return _length;
153 }
154
155 // Whether the array is empty
156 bool is_empty() const {

```



```

149     return _length == 0;
150 }
151
152 // Current capacity
153 size_t capacity() const {
154     return _capacity;
155 }
156
157 void clear_and_reserve(size_t new_capacity) {
158     _length = 0;
159     if (_data != nullptr) {
160         FreeHeap(_data);
161     }
162     _capacity = MAX2((size_t) 8, next_power_of_2(
163         new_capacity - 1));
164     _data = (ZWeakRefData*)AllocateHeap(_capacity *
165         sizeof(ZWeakRefData), mtGC);
166 }
167
168 // Clear without deallocating
169 void clear() {
170     _length = 0;
171 }
172
173 // Reserve capacity without clearing
174 void reserve(size_t new_capacity) {
175     if (new_capacity > _capacity) {
176         grow(new_capacity);
177     }
178 }
179 };
180
181 #endif // SHARE_GC_Z_ZADDRESSARRAY_HPP

```

Reference Processor Interface

File: src/hotspot/share/gc/z/zReferenceProcessor.hpp

```

1  /*
2   * Copyright (c) 2015, 2024, Oracle and/or its
3   * affiliates. All rights reserved.
4   * DO NOT ALTER OR REMOVE COPYRIGHT NOTICES OR THIS FILE
5   * HEADER.
6   *
7   * This code is free software; you can redistribute it
8   * and/or modify it

```

```

6  * under the terms of the GNU General Public License
   * version 2 only, as
7  * published by the Free Software Foundation.
8  *
9  * This code is distributed in the hope that it will be
   * useful, but WITHOUT
10 * ANY WARRANTY; without even the implied warranty of
   * MERCHANTABILITY or
11 * FITNESS FOR A PARTICULAR PURPOSE. See the GNU
   * General Public License
12 * version 2 for more details (a copy is included in the
   * LICENSE file that
13 * accompanied this code).
14 *
15 * You should have received a copy of the GNU General
   * Public License version
16 * 2 along with this work; if not, write to the Free
   * Software Foundation,
17 * Inc., 51 Franklin St, Fifth Floor, Boston, MA
   * 02110-1301 USA.
18 *
19 * Please contact Oracle, 500 Oracle Parkway, Redwood
   * Shores, CA 94065 USA
20 * or visit www.oracle.com if you need additional
   * information or have any
21 * questions.
22 */
23
24 #ifndef SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP
25 #define SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP
26
27 #include "gc/shared/referenceDiscoverer.hpp"
28 #include "gc/z/zAddress.hpp"
29 #include "gc/z/zAddressArray.hpp"
30 #include "gc/z/zValue.hpp"
31 #include "gc/z/zConfFlags.h"
32
33 class ConcurrentGCTimer;
34 class ReferencePolicy;
35 class ZWorkers;
36
37
38
39 class ZReferenceProcessor : public ReferenceDiscoverer {
40     friend class ZReferenceProcessorTask;
41
42 private:

```

```

43  static const size_t ReferenceTypeCount = REF_PHANTOM +
    1;
44  typedef size_t Counters[ReferenceTypeCount];
45
46  ZWorkers* const _workers;
47  ReferencePolicy* _soft_reference_policy;
48  bool
    _uses_clear_all_soft_reference_policy;
49  ZPerWorker<Counters> _encountered_count;
50  ZPerWorker<Counters> _discovered_count;
51  ZPerWorker<Counters> _enqueued_count;
52  ZPerWorker<size_t>
    _encountered_weak_refs_without_queue_count;
53  ZPerWorker<size_t>
    _discovered_weak_refs_without_queue_count;
54  ZPerWorker<size_t>
    _cleared_weak_refs_without_queue_count;
55  ZPerWorker<zaddress> _discovered_list;
56  ZContented<zaddress> _pending_list;
57  zaddress _pending_list_tail;
58  #if ZUseSeparateDiscoveredLists
59  #if ZUseDynamicArray
60      ZPerWorker<ZAddressArray>
        _discovered_weak_refs_without_queue_arr;
61      ZPerWorker<bool> _array_empty;
62  #else // !ZUseDynamicArray
63      ZPerWorker<zaddress>
        _discovered_weak_refs_without_queue_ll;
64  #endif // ZUseDynamicArray
65  #elif ZUseDynamicArray // !ZUseSeparateDiscoveredLists
        && ZUseDynamicArray
66      ZPerWorker<ZAddressArray> _discovered_all_refs_arr;
67      ZPerWorker<bool> _all_refs_array_empty;
68  #endif // ZUseSeparateDiscoveredLists
69      OopHandle _null_queue_handle;
70
71      bool is_inactive(zaddress reference, oop referent,
        ReferenceType type) const;
72      bool is_strongly_live(oop referent) const;
73      bool is_softly_live(zaddress reference, ReferenceType
        type) const;
74
75      bool should_discover(zaddress reference, ReferenceType
        type, oop referent) const;
76      bool try_make_inactive(zaddress reference,
        ReferenceType type) const;
77  #if ZUseOptimisedClearPath

```

```

78     bool try_make_inactive_fast(const ZWeakRefData& data);
79 #endif // ZUseOptimisedClearPath
80
81     void discover(zaddress reference, ReferenceType type,
82                 zaddress referent);
83
84     void verify_empty() const;
85
86     void process_worker_discovered_list(zaddress
87         discovered_list);
88 #if ZUseSeparateDiscoveredLists && ZUseDynamicArray
89     void process_worker_discovered_weak_refs_without_queue
90         (ZAddressArray& weak_refs_without_queue);
91 #elif ZUseSeparateDiscoveredLists && !ZUseDynamicArray
92     void process_worker_discovered_weak_refs_without_queue
93         (zaddress weak_refs_without_queue);
94 #elif !ZUseSeparateDiscoveredLists && ZUseDynamicArray
95     void process_worker_discovered_all_refs(ZAddressArray&
96         all_refs);
97 #endif // ZUseSeparateDiscoveredLists &&
98         ZUseDynamicArray
99     void work();
100     void collect_statistics();
101
102     zaddress swap_pending_list(zaddress pending_list);
103
104     inline bool has_reference_queue(zaddress reference);
105
106     void initialize_null_queue_handle();
107
108 public:
109     ZReferenceProcessor(ZWorkers* workers);
110
111     void set_soft_reference_policy(bool
112         clear_all_soft_references);
113     bool uses_clear_all_soft_reference_policy() const;
114
115     void reset_statistics();
116
117     virtual bool discover_reference(oop reference,
118         ReferenceType type);
119     void process_references();
120     void enqueue_references();
121
122     void verify_pending_references();
123
124     void prepare();

```

```

117 };
118
119 #endif // SHARE_GC_Z_ZREFERENCEPROCESSOR_HPP

```

Reference Processor Implementation (Excerpts)

Excerpt: `src/hotspot/share/gc/z/zReferenceProcessor.cpp` (constructor and per-worker storage setup)

```

1   head = reference;
2   } else {
3       // Not first append, link tail
4       reference_set_discovered(tail, reference);
5   }
6
7   // Always set tail
8   tail = reference;
9   }
10
11 ZReferenceProcessor::ZReferenceProcessor(ZWorkers*
    workers)
12 : _workers(workers),
13   _soft_reference_policy(nullptr),
14   _uses_clear_all_soft_reference_policy(false),
15   _encountered_count(),
16   _discovered_count(),
17   _enqueued_count(),
18   _encountered_weak_refs_without_queue_count(),
19   _discovered_weak_refs_without_queue_count(),
20   _cleared_weak_refs_without_queue_count(),
21   _discovered_list(),
22   _pending_list(zaddress::null),
23   _pending_list_tail(zaddress::null),
24   #if ZUseSeparateDiscoveredLists
25   #if ZUseDynamicArray
26       _discovered_weak_refs_without_queue_arr(),
27       _array_empty(),
28   #else // !ZUseDynamicArray
29       _discovered_weak_refs_without_queue_ll(),
30   #endif // ZUseDynamicArray
31   #elif ZUseDynamicArray // !ZUseSeparateDiscoveredLists
32       && ZUseDynamicArray
33       _discovered_all_refs_arr(),
34       _all_refs_array_empty(),
35   #endif // ZUseSeparateDiscoveredLists
36   _null_queue_handle() {

```

```

36 #if ZUseSeparateDiscoveredLists && ZUseDynamicArray
37     _array_empty.set_all(true);
38 #elif ZUseDynamicArray // !ZUseSeparateDiscoveredLists
39     && ZUseDynamicArray
40     _all_refs_array_empty.set_all(true);
41 #endif
42
43     log_info(gc, ref) ("Reference Processor created, Macros
44         : ZUseSeparateDiscoveredLists=%d, ZUseDynamicArray
45         =%d, ZUseOptimisedClearPath=%d",
46         ZUseSeparateDiscoveredLists, ZUseDynamicArray,
47         ZUseOptimisedClearPath);
48 }
49
50 void ZReferenceProcessor::set_soft_reference_policy(bool
51     clear_all_soft_references) {
52     static AlwaysClearPolicy always_clear_policy;
53     static LRUMaxHeapPolicy lru_max_heap_policy;
54
55     _uses_clear_all_soft_reference_policy =
56         clear_all_soft_references;
57
58     if (clear_all_soft_references) {
59         _soft_reference_policy = &always_clear_policy;
60     } else {
61         _soft_reference_policy = &lru_max_heap_policy;
62     }
63
64     _soft_reference_policy->setup();
65 }
66
67 bool ZReferenceProcessor::
68     uses_clear_all_soft_reference_policy() const {
69     return _uses_clear_all_soft_reference_policy;
70 }
71
72 bool ZReferenceProcessor::is_inactive(zaddress reference
73     , oop referent, ReferenceType type) const {
74     if (type == REF_FINAL) {
75         // A FinalReference is inactive if its next field is
76         // non-null. An application can't
77         // call enqueue() or clear() on a FinalReference.
78         return !is_null(reference_next(reference));
79     } else {
80         // Verification
81         check_is_valid_zaddress(referent);

```

Excerpt: src/hotspot/share/gc/z/zReferenceProcessor.cpp (discovery path with queue-less weak-reference routing)

```
1 void ZReferenceProcessor::discover(zaddress reference,
   ReferenceType type, zaddress referent) {
2   log_trace(gc, ref)("Discovered Reference: " PTR_FORMAT
   " (%s)", untype(reference), reference_type_name(
   type));
3
4   // Update statistics
5   if (type == REF_WEAK && !has_reference_queue(reference)
   ) {
6     _discovered_weak_refs_without_queue_count.get()++;
7   }
8   else {
9     _discovered_count.get()[type]++;
10  }
11
12  assert(ZHeap::heap()->is_old(reference), "Must be old"
   );
13  assert(is_null(reference_discovered(reference)), "
   Already discovered");
14
15  #if ZUseSeparateDiscoveredLists
16    if (type == REF_WEAK && !has_reference_queue(reference)
   ) {
17      #if ZUseDynamicArray && ZUseOptimisedClearPath
18        zpointer* const referent_addr =
19          reference_referent_addr_non_vol(reference);
20        zaddress* const discovered_addr =
21          reference_discovered_addr(reference);
22        const zpointer referent_value = *referent_addr;
23
24        // WeakReference with null ReferenceQueue - remember
25        // for special processing
26        ZAddressArray& weak_refs_without_queue =
27          _discovered_weak_refs_without_queue_arr.get();
28        weak_refs_without_queue.append(referent_addr,
29          discovered_addr,
30          referent,
31          referent_value);
32
33        _array_empty.set(false);
34        reference_set_discovered(reference, reference); //
35        mark as discovered
36      #elif ZUseDynamicArray && !ZUseOptimisedClearPath
37        // WeakReference with null ReferenceQueue - remember
38        // for special processing
```

```

32     ZAddressArray& weak_refs_without_queue =
        _discovered_weak_refs_without_queue_arr.get();
33     weak_refs_without_queue.append(reference);
34     _array_empty.set(false);
35     reference_set_discovered(reference, reference); //
        mark as discovered
36 #else // !ZUseDynamicArray
37     zaddress* const head =
        _discovered_weak_refs_without_queue_ll.addr();
38     reference_set_discovered(reference, *head);
39     *head = reference;
40 #endif // ZUseDynamicArray
41     } else {
42         if (type == REF_FINAL) {
43             // Mark referent (and its reachable subgraph)
                finalizable. This avoids
44             // the problem of later having to mark those
                objects if the referent is
45             // still final reachable during processing.
46             volatile zpointer* const referent_addr =
                reference_referent_addr(reference);
47             ZBarrier::mark_barrier_on_old_oop_field(
                referent_addr, true /* finalizable */);
48         }
49
50         // Add reference to discovered list
51         zaddress* const head = _discovered_list.addr();
52         reference_set_discovered(reference, *head);
53         *head = reference;
54     }
55 #elif !ZUseSeparateDiscoveredLists && ZUseDynamicArray
56     if (type == REF_FINAL) {
57         // Mark referent (and its reachable subgraph)
            finalizable. This avoids
58         // the problem of later having to mark those objects
            if the referent is
59         // still final reachable during processing.
60         volatile zpointer* const referent_addr =
            reference_referent_addr(reference);
61         ZBarrier::mark_barrier_on_old_oop_field(
            referent_addr, true /* finalizable */);
62     }
63
64     // Add all reference types to a single dynamic array
        for processing
65     {

```



```

66     ZAddressArray& all_refs = _discovered_all_refs_arr.
        get();
67 #if ZUseOptimisedClearPath
68     zpointer* const referent_field =
        reference_referent_addr_non_vol(reference);
69     zaddress* const discovered_field =
        reference_discovered_addr(reference);
70     const zpointer referent_value = *referent_field;
71     all_refs.append(referent_field, discovered_field,
        referent, referent_value, reference);
72 #else // !ZUseOptimisedClearPath
73     all_refs.append(reference);
74 #endif // ZUseOptimisedClearPath
75     _all_refs_array_empty.set(false);
76     reference_set_discovered(reference, reference); //
        mark as discovered
77 }
78 #else // !ZUseSeparateDiscoveredLists && !
        ZUseDynamicArray
79     if (type == REF_FINAL) {
80         // Mark referent (and its reachable subgraph)
            finalizable. This avoids
81         // the problem of later having to mark those objects
            if the referent is
82         // still final reachable during processing.
83         volatile zpointer* const referent_addr =
            reference_referent_addr(reference);
84         ZBarrier::mark_barrier_on_old_oop_field(
            referent_addr, true /* finalizable */);
85     }
86
87     // Add reference to discovered list
88     zaddress* const head = _discovered_list.addr();
89     reference_set_discovered(reference, *head);
90     *head = reference;
91 #endif // ZUseSeparateDiscoveredLists

```

Excerpt: `src/hotspot/share/gc/z/zReferenceProcessor.cpp` (processing paths for separate lists and dynamic-array variants)

```

1     if (is_null(old_pending_list)) {
2         // Old list was empty. First to prepend to list,
            record tail
3         _pending_list_tail = keep_tail;
4     } else {
5         assert(ZHeap::heap()->is_old(old_pending_list), "
            Must be old");
6     }

```

```

7     }
8 }
9 #if ZUseSeparateDiscoveredLists && ZUseDynamicArray
10 void ZReferenceProcessor::
    process_worker_discovered_weak_refs_without_queue(
        ZAddressArray& weak_refs_without_queue) {
11     size_t dropped = 0;
12     for (size_t i = 0; i < weak_refs_without_queue.length
        ()); i++) {
13         const ZWeakRefData& data = weak_refs_without_queue.
            at(i);
14 #if ZUseOptimisedClearPath
15         *data.discovered_field_addr = zaddress::null; //
            Mark as dropped
16         if (try_make_inactive_fast(data)) {
17             log_trace(gc, ref) ("\nEnqueued\n Weak Reference
                without Queue");
18             // Update statistics
19             _cleared_weak_refs_without_queue_count.get()++;
20         } else {
21             log_trace(gc, ref) ("Dropped Weak Reference without
                Queue");
22             dropped++;
23         }
24 #else // !ZUseOptimisedClearPath
25         zaddress reference = data.reference;
26         reference_set_discovered(reference, zaddress::null);
27         // Mark as dropped
28         if (try_make_inactive(reference, reference_type(
            reference))) {
29             log_trace(gc, ref) ("\nEnqueued\n Weak Reference
                without Queue");
30             // Update statistics
31             _cleared_weak_refs_without_queue_count.get()++;
32         } else {
33             log_trace(gc, ref) ("Dropped Weak Reference without
                Queue");
34             dropped++;
35         }
36 #endif // ZUseOptimisedClearPath
37         SuspendibleThreadSet::yield();
38     }
39     weak_refs_without_queue.clear_and_reserve(dropped);
40 }
41 #elif ZUseSeparateDiscoveredLists && !ZUseDynamicArray

```

```

41 void ZReferenceProcessor::
    process_worker_discovered_weak_refs_without_queue(
        zaddress weak_refs_without_queue) {
42     for (zaddress current = weak_refs_without_queue; !
        is_null(current);) {
43         zaddress next = reference_discovered(current);
44     #if ZUseOptimisedClearPath
45         ZWeakRefData data;
46         data.referent_field_addr =
            reference_referent_addr_non_vol(current);
47         data.discovered_field_addr =
            reference_discovered_addr(current);
48         data.referent_field_value = *data.
            referent_field_addr;
49         data.referent_addr = ZBarrier::
            load_barrier_on_oop_field(reference_referent_addr
                (current));
50         *data.discovered_field_addr = zaddress::null; //
            Mark as dropped
51         if (try_make_inactive_fast(data)) {
52             log_trace(gc, ref) ("\nEnqueued\n Weak Reference
                without Queue");
53             // Update statistics
54             _cleared_weak_refs_without_queue_count.get()++;
55         } else {
56             log_trace(gc, ref) ("Dropped Weak Reference without
                Queue");
57         }
58     #else //!ZUseOptimisedClearPath
59         reference_set_discovered(current, zaddress::null);
60         // Mark as dropped
61         if (try_make_inactive(current, reference_type(
            current))) {
62             log_trace(gc, ref) ("\nEnqueued\n Weak Reference
                without Queue");
63             // Update statistics
64             _cleared_weak_refs_without_queue_count.get()++;
65         } else {
66             log_trace(gc, ref) ("Dropped Weak Reference without
                Queue");
67         }
68     #endif // ZUseOptimisedClearPath
69     current = next;
70     SuspendibleThreadSet::yield();
71 }
72 #elif !ZUseSeparateDiscoveredLists && ZUseDynamicArray

```

```

73 void ZReferenceProcessor::
    process_worker_discovered_all_refs(ZAddressArray&
    all_refs) {
74     zaddress keep_head = zaddress::null;
75     zaddress keep_tail = zaddress::null;
76     size_t dropped = 0;
77
78     for (size_t i = 0; i < all_refs.length(); i++) {
79         const ZWeakRefData& data = all_refs.at(i);
80         const zaddress reference = data.reference;
81         const ReferenceType type = reference_type(reference)
            ;
82         const bool is_weak_no_queue = (type == REF_WEAK && !
            has_reference_queue(reference));
83
84         if (is_weak_no_queue) {
85             #if ZUseOptimisedClearPath
86                 *data.discovered_field_addr = zaddress::null;
87                 if (try_make_inactive_fast(data)) {
88                     log_trace(gc, ref) ("\"Enqueued\" Weak Reference
                        without Queue");
89                     _cleared_weak_refs_without_queue_count.get()++;
90                 } else {
91                     log_trace(gc, ref) ("Dropped Weak Reference
                        without Queue");
92                     dropped++;
93                 }
94             #else // !ZUseOptimisedClearPath
95                 reference_set_discovered(reference, zaddress::null
                    );
96                 if (try_make_inactive(reference, type)) {
97                     log_trace(gc, ref) ("\"Enqueued\" Weak Reference
                        without Queue");
98                     _cleared_weak_refs_without_queue_count.get()++;
99                 } else {
100                     log_trace(gc, ref) ("Dropped Weak Reference
                        without Queue");
101                     dropped++;
102                 }
103             #endif // ZUseOptimisedClearPath
104         } else {
105             reference_set_discovered(reference, zaddress::null
                );
106             if (try_make_inactive(reference, type)) {
107                 log_trace(gc, ref) ("Enqueued Reference: "
                    PTR_FORMAT " (%s)", untype(reference),
                    reference_type_name(type));

```

```

108     _enqueued_count.get()[type]++;
109     list_append(keep_head, keep_tail, reference);
110 } else {
111     log_trace(gc, ref)("Dropped Reference: "
112         PTR_FORMAT " (%s)", untype(reference),
113         reference_type_name(type));
114     dropped++;
115 }
116 }
117 }
118
119 SuspensibleThreadSet::yield();
120
121 all_refs.clear_and_reserve(dropped);
122
123 // Prepend kept references to the internal pending
124 // list
125 if (!is_null(keep_head)) {
126     const zaddress old_pending_list = AtomicAccess::xchg
127         (_pending_list.addr(), keep_head);
128     reference_set_discovered(keep_tail, old_pending_list);
129 }
130 if (is_null(old_pending_list)) {
131     _pending_list_tail = keep_tail;
132 } else {
133     assert(ZHeap::heap()->is_old(old_pending_list), "
134         Must be old");
135 }
136 }
137 }
138 #endif // ZUseSeparateDiscoveredLists &&
139 ZUseDynamicArray
140
141 void ZReferenceProcessor::work() {
142     SuspensibleThreadSetJoiner sts_joiner;
143
144 #if ZUseSeparateDiscoveredLists
145 #if ZUseDynamicArray
146     ZPerWorkerIterator<zaddress> iter(&_discovered_list);
147     ZPerWorkerIterator<ZAddressArray> iter_weak_refs(&
148         _discovered_weak_refs_without_queue_arr);
149     ZPerWorkerIterator<bool> iter_array_empty(&
150         _array_empty);
151
152     zaddress* list_addr = nullptr;
153     ZAddressArray* array_addr = nullptr;
154     bool* array_empty = nullptr;

```

```

146
147 for (; iter.next(&list_addr) && iter_weak_refs.next(&
    array_addr) && iter_array_empty.next(&array_empty)
    ;) {
148
149     const zaddress discovered_list = AtomicAccess::xchg(
        list_addr, zaddress::null);
150     const bool has_array = !AtomicAccess::xchg(
        array_empty, true);
151     const bool has_discovered = discovered_list !=
        zaddress::null;
152
153     if (has_discovered) {
154         process_worker_discovered_list(discovered_list);
155     }
156     if (has_array) {
157         process_worker_discovered_weak_refs_without_queue
            (*array_addr);
158     }
159 }
160 #else // !ZUseDynamicArray
161     ZPerWorkerIterator<zaddress> iter(&_discovered_list);
162     ZPerWorkerIterator<zaddress> iter_weak_refs(&
        _discovered_weak_refs_without_queue_ll);
163
164     zaddress* list_addr = nullptr;
165     zaddress* no_queue_list_addr = nullptr;
166
167     for (; iter.next(&list_addr) && iter_weak_refs.next(&
        no_queue_list_addr);) {
168         const zaddress discovered_list = AtomicAccess::xchg(
            list_addr, zaddress::null);
169         const zaddress weak_refs_without_queue =
            AtomicAccess::xchg(no_queue_list_addr, zaddress::
                null);
170
171         if (!is_null(discovered_list)) {
172             process_worker_discovered_list(discovered_list);
173         }
174         if (!is_null(weak_refs_without_queue)) {
175             process_worker_discovered_weak_refs_without_queue(
                weak_refs_without_queue);
176         }
177     }
178 #endif // ZUseDynamicArray
179 #else // !ZUseSeparateDiscoveredLists
180 #if ZUseDynamicArray

```

```

181 ZPerWorkerIterator<ZAddressArray> iter_all_refs(&
    _discovered_all_refs_arr);
182 ZPerWorkerIterator<bool> iter_all_refs_empty(&
    _all_refs_array_empty);
183
184 ZAddressArray* all_refs_addr = nullptr;
185 bool* all_refs_empty = nullptr;
186
187 for (; iter_all_refs.next(&all_refs_addr) &&
    iter_all_refs_empty.next(&all_refs_empty);) {
188     const bool has_array = !AtomicAccess::xchg(
        all_refs_empty, true);
189     if (has_array) {
190         process_worker_discovered_all_refs(*all_refs_addr)
            ;
191     }
192 }
193 #else // !ZUseDynamicArray
194 ZPerWorkerIterator<zaddress> iter(&_discovered_list);
195 for (zaddress* list_addr; iter.next(&list_addr);) {
196     const zaddress discovered_list = AtomicAccess::xchg(
        list_addr, zaddress::null);
197     if (!is_null(discovered_list)) {
198         process_worker_discovered_list(discovered_list);
199     }
200 }
201 #endif // ZUseDynamicArray
202 #endif // ZUseSeparateDiscoveredLists
203 }
204
205 void ZReferenceProcessor::verify_empty() const {
206 #ifdef ASSERT

```

Excerpt: src/hotspot/share/gc/z/zReferenceProcessor.cpp (queue sentinel initialisation and queue checks)

```

1
2     // Notify ReferenceHandler thread
3     ml.notify_all();
4 }
5
6     // Reset internal pending list
7     _pending_list.set(zaddress::null);
8     _pending_list_tail = zaddress::null;
9 }
10
11 inline bool ZReferenceProcessor::has_reference_queue(
    zaddress reference) {

```

```

12   oop ref_queue = to_oop(reference)->obj_field_access<
      AS_NO_KEEPLIVE>(java_lang_ref_Reference::
      queue_offset());
13   bool result = ref_queue != _null_queue_handle.resolve
      ();
14   return result;
15 }
16
17 void ZReferenceProcessor::initialize_null_queue_handle()
    {
18     EXCEPTION_MARK;
19     TempNewSymbol class_name = SymbolTable::new_symbol("
      java/lang/ref/ReferenceQueue");
20     Klass* k = SystemDictionary::resolve_or_fail(
      class_name, true, CHECK);
21     InstanceKlass* ik = InstanceKlass::cast(k);
22     ik->initialize(CHECK);
23     fieldDescriptor fd;
24     bool found = ik->find_local_field(SymbolTable::
      new_symbol("NULL_QUEUE"),
25                                     vmSymbols::
      referencequeue_signature
      (), &fd);
26     assert(found && fd.is_static(), "ReferenceQueue.
      NULL_QUEUE missing");
27     oop null_q = ik->java_mirror()->obj_field(fd.offset())
      ;
28     _null_queue_handle = OopHandle(Universe::vm_global(),
      null_q);
29 }
30
31 void ZReferenceProcessor::prepare() {
32     initialize_null_queue_handle();
33 }

```